# BACHELORTHESIS

## In Biomedical Engineering

# BOINSO – An open source solution for distributed ground and mission control communication for low earth orbit satellites

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# Declaration

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# Kurzfassung

In den letzten Jahren besteht die Möglichkeit für Forschungseinrichtungen und somit für die Lehre verwendete, kleine Satelliten zu bauen und in den Orbit zu schicken. Ermöglicht wurde dies durch die 1999 an der California Polytechnic State University (Cal Poly) entwickelten CubeSats.
Da CubeSats sich im Low Earth Orbit (LEO) befinden, besteht teilweise nur ein 15 minütiges Zeitfenster in dem der Satellit das Mission Control Center (MCC) passiert, welches zu einer sehr geringen Datenausbeute führt [Preindl, 2010]. Um mehr Daten zu erhalten, wurde Global Educational Network for Satellite Operations (GENSO) entwickelt, ein weltweites Netzwerk aus Bodenstationen welche nun nichtmehr nur auf deren eigene CubeSats sondern, mit Berechtigung der MCCs, auch auf andere zugreifen können [Leveque et al., 2007].
Diese Arbeit beschäftigt sich mit der Entwicklung einer neuen Open Source Alternative, Berkley Open Infrastructure for Networking Satellite Operations (BOINSO), welche mithilfe von bereits vorhandenen Open Source Projekten und der Infrastruktur der Berkeley Open Infrastructure for Network Computing (BOINC) realisiert wird [Ries, 2012].
BOINSO wird unter der GNU General Public License (GNU GPL) vertrieben und steht unter https://github.com/Mehnen/BOINSO zum Download bereit.

**Schlagworte:** CubeSat, GENSO, BOINSO

# Abstract

During the last few years, there was a major increase in educational satellites launched to the orbit. The start of this was in 1999, when scientists at California Polytechnic State University (Cal Poly) developed the CubeSats.
Because of the CubeSats being in LEO, there is a very small timeframe for the satellites to send data to the MCC which leads to a very small data volume [Preindl, 2010]. To be able to get more data GENSO was devised which connects a network of Ground Control Centers (GCCs) all over the world. Those GCCs can access the CubeSat of other teams and transmit the data of the passing satellite [Leveque et al., 2007].
This thesis focuses on the development of a new open source alternative, BOINSO, which is build using existing open source projects and the interface of BOINC [Ries, 2012].
BOINSO is released under the GNU GPL and is available at https://github.com/Mehnen/BOINSO.

**Keywords:** CubeSat, GENSO, BOINSO

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# 1 Introduction

Educational satellites like the CubeSats reside almost exclusively at the LEO, which ranges from 200 km to 2000 km above the earth surface, which has various reasons.
CubeSats are very small satellites and therefore the space for electronic devices and accumulators is very limited. Thus the short range between the device and GCCs is ideal as it reduces the power consumtion to a minimum and is close enough to earth so that the CubeSat will not be affected by the radiation of the Van Allen radiation belts [Horne et al., 2005].
Moreover the deployment of a satellites in LEO significantly decreases the costs compared to satellites stationed in the Medium Earth Orbit (MEO) or Geostationary Orbit (GEO).
Furthermore, the closer the small satellites are to earth, the sooner they will burn to ashes caused by entering earth's atmosphere. This helps keeping the orbits clean, as space debris can cause collateral damage to future space missions [Akahoshi et al., 2008].
However, the small distance a satellite has in LEO leads to a severe problem to retrieve data, as data transmission is only possible when the CubeSat is visible to the GCC, leading to communication time-frames as short as 30 minutes. This fact, combined with a slow data rate makes a program like BOINSO indispensable.

# 2 Background

## 2.1 GENSO

GENSO's aim is to provide a transmission network enabling a MCC to communicate with their respective educational satellites the whole day via GCCs located around the globe. This would prevent GCCs all over the world to reside unused for the major time of the day as normally every CubeSat sends to only one station.
Another aspect which

# 3 Glossary

**GENSO** Global Educational Network for Satellite Operations

**Cal Poly** California Polytechnic State University

**LEO** Low Earth Orbit

**MCC** Mission Control Center

**BOINSO** Berkley Open Infrastructure for Networking Satellite Operations

**GNU GPL** GNU General Public License

**GCC** Ground Control Center

**BOINC** Berkeley Open Infrastructure for Network Computing

**MEO** Medium Earth Orbit

**GEO** Geostationary Orbit

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